

Commodity vs. Equity Options Pricing

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1. Underlying Assets

Denote time to maturity $\tau = T - t$.

- **Commodity Options:** Underlying is futures price $F(t, T)$, linked to spot S_t by cost-of-carry:

$$F(t, T) = S_t \exp[(r + u - y)\tau],$$

where r = risk-free rate, u = storage cost rate, y = convenience yield.

- **Equity Options:** Underlying is spot S_t with continuous dividend yield q :

$$F_{eq}(t, T) = S_t \exp[(r - q)\tau].$$

2. Cost-of-Carry & Convenience Yield

From forward definitions:

$$F_{comm} = S_t \exp[(r + u - y)\tau],$$
$$F_{eq} = S_t \exp[(r - q)\tau].$$

Solve for convenience yield y :

$$y = r + u - \frac{1}{\tau} \ln \frac{F_{comm}}{S_t}.$$

The convenience yield y acts like a negative dividend, capturing non-monetary benefits of holding inventory.

3. Stochastic Dynamics

Commodity Spot (Ornstein–Uhlenbeck):

$$dS_t = \kappa(\theta - S_t) dt + \sigma dW_t,$$

where κ = speed of mean reversion, θ = long-run level.

(Alternatively, model log-spot OU: $d \ln S_t = \kappa(\mu - \ln S_t)dt + \sigma dW_t$.)

Equity Spot (GBM):

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t.$$

Under risk-neutral measure, drift is $r - q$ and volatility σ implies lognormal S_t .

4. Valuation Models

Black's Model (Futures Options):

$$C_{comm} = e^{-r\tau} [FN(d_1) - KN(d_2)],$$
$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Pricing applies directly to futures as underlyings under lognormal assumption.

Black–Scholes–Merton (Equity):

$$C_{eq} = S_t e^{-q\tau} N(d_1) - K e^{-r\tau} N(d_2),$$
$$d_1 = \frac{\ln(S_t/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}},$$

where $d_2 = d_1 - \sigma\sqrt{\tau}$ and $F_{eq} = S_t e^{(r-q)\tau}$ imbedded in d_i .

5. Volatility Term Structures

Commodities: Implied vol $\sigma_{imp}(\tau)$ depends on forward curve shape (contango/backwardation) and often exhibits hump or inversion under backwardation.

Equities: Implied vol surface shows skew/smile:

$$\sigma_{imp}(K, \tau) \approx \sigma_0 + a \ln \frac{K}{F_{eq}} + b\tau,$$

with parameters a, b calibrated to capture strike and term effects from leverage and risk premium.

6. Exercise Styles Market Liquidity

- **Exercise Style:** Commodity options are typically European—no early exercise premium. Many equity options (single stock) are American, requiring binomial/PDE for early exercise valuation.
- **Liquidity:** OTC commodity options often have wider bid-ask spreads and lower depth, raising execution costs and implied vol calibration. Equity options on large caps/indices are highly liquid with tight spreads.

- Commodity options: require modeling of carry, convenience yield, OU dynamics, and futures-based Black pricing.
- Equity options: rely on GBM, dividend yield adjustments, skewed vol surfaces and early exercise.
- Accurate valuation and hedging depend on choosing the correct SDE, cost-of-carry framework, and pricing model per market.